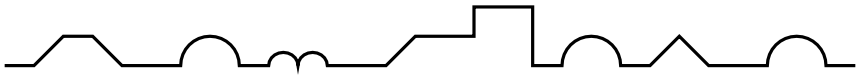
Math Kangaroo Practice

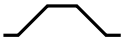
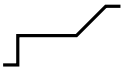
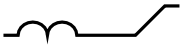
Grade 3-4 · 24 questions · harder · seed 94478

Part A: Each question is worth 3 points

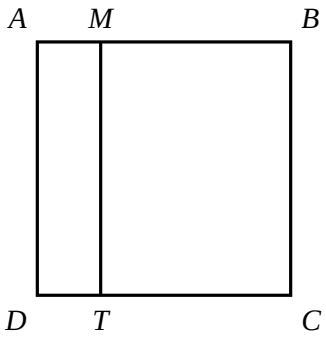
1. Far away we see the skyline of a castle.



Which of the pieces **cannot** belong to the skyline?

A.  ***B.*** ***C.***  ***D.***  ***E.***

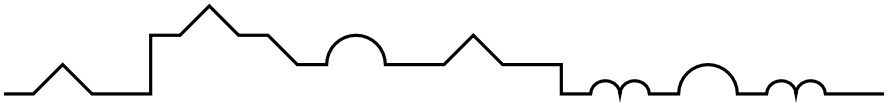
2. *ABCD* is a square. Its side is 20 cm. The line *MT* cuts it into two rectangles, and *AM* = 5 cm.



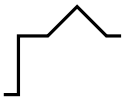
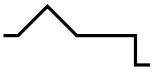
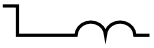
How many centimetres is the perimeter of the square *ABCD* larger than that of the rectangle *MBCT*?

A. 70 cm ***B.*** 40 cm ***C.*** 30 cm ***D.*** 20 cm ***E.*** 10 cm

3. Lena cut a strip of paper along this line.

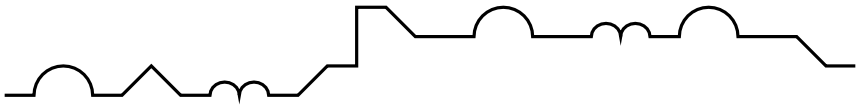


Which piece **cannot** come from the cut line?

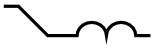
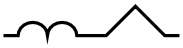
A. ***B.***  ***C.*** ***D.***  ***E.*** 

Part B: Each question is worth 4 points

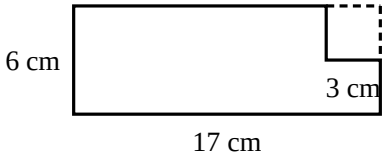
4. Far away we see the skyline of a castle.



Which of the pieces **can** belong to the skyline?

A. ***B.***  ***C.*** ***D.***  ***E.***

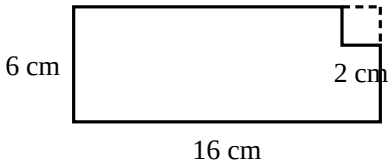
5. A rectangle is 17 cm long and 6 cm wide. A square with side 3 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 46 cm ***B.*** 50 cm ***C.*** 52 cm ***D.*** 53 cm ***E.*** 58 cm

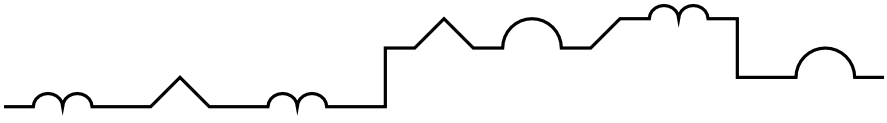
6. A rectangle is 16 cm long and 6 cm wide. A square with side 2 cm is cut off one corner.



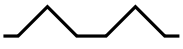
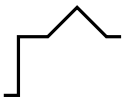
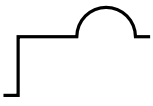
What is the perimeter of the remaining shape?

A. 40 cm ***B.*** 42 cm ***C.*** 44 cm ***D.*** 48 cm ***E.*** 52 cm

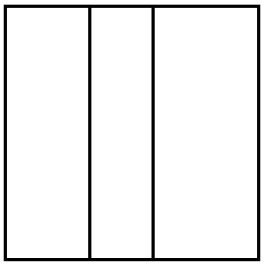
7. Lena cut a strip of paper along this line.



Which piece **can** come from the cut line?

A.  ***B.***  ***C.*** ***D.***  ***E.***

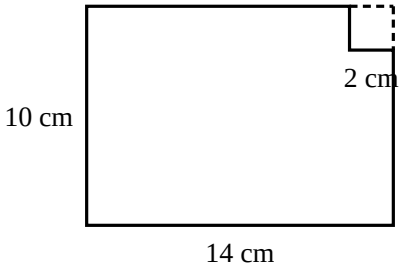
8. A square garden is surrounded by a fence 36 m long. The gardener divides it into 3 rectangular beds as shown and puts a separate fence around each bed.



How many metres of fence are around the beds altogether?

A. 72 m ***B.*** 63 m ***C.*** 62 m ***D.*** 54 m ***E.*** 36 m

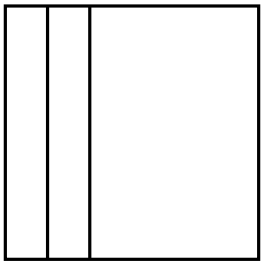
9. A rectangle is 14 cm long and 10 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 48 cm ***B.*** 46 cm ***C.*** 44 cm ***D.*** 40 cm ***E.*** 24 cm

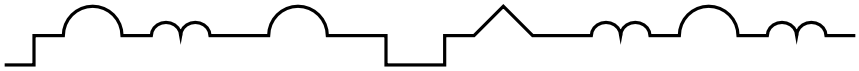
10. A square garden is surrounded by a fence 40 m long. The gardener divides it into 3 rectangular beds as shown and puts a separate fence around each bed.



How many metres of fence are around the beds altogether?

A. 120 m ***B.*** 89 m ***C.*** 80 m ***D.*** 60 m ***E.*** 40 m

11. This is the outline of a town seen from the river at sunset.

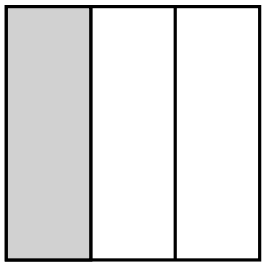


Which piece **can** be part of the outline?

A. ***B.*** ***C.*** ***D.*** ***E.***

Part C: Each question is worth 5 points

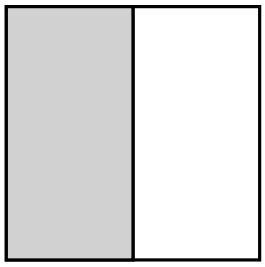
12. Mia cuts a square photo into 3 equal strips as shown. The perimeter of each strip is 56 cm.



What is the perimeter of the square?

A. 168 cm ***B.*** 112 cm ***C.*** 101 cm ***D.*** 88 cm ***E.*** 84 cm

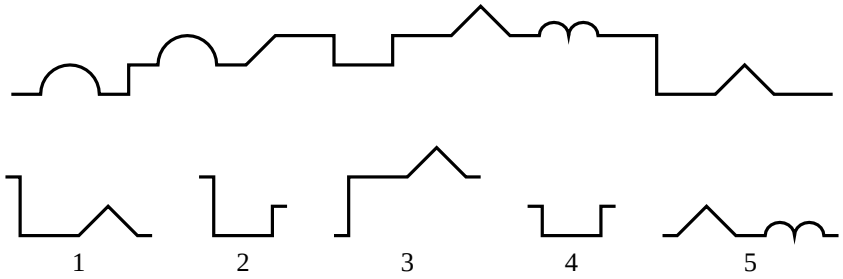
13. Mia cuts a square photo into 2 equal strips as shown. The perimeter of each strip is 54 cm.



What is the perimeter of the square?

A. 108 cm ***B.*** 81 cm ***C.*** 75 cm ***D.*** 74 cm ***E.*** 72 cm

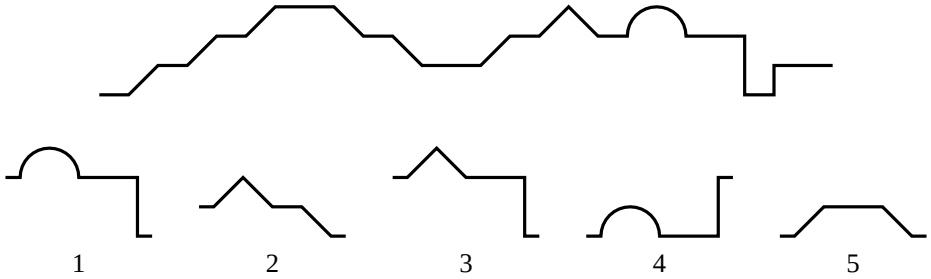
14. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 3 ***D.*** 4 ***E.*** 5

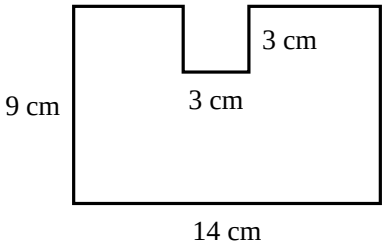
15. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

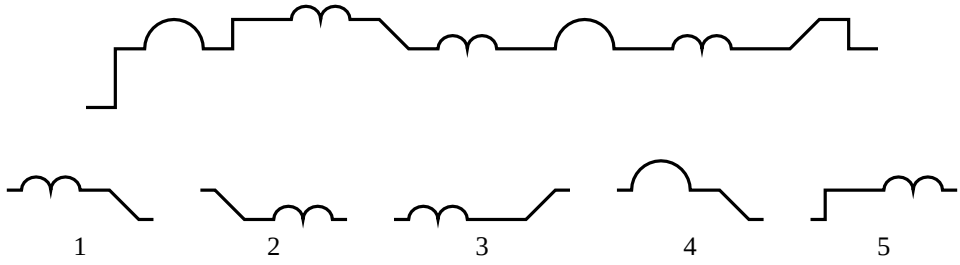
16. A 14 cm by 9 cm rectangle of cardboard has a rectangular notch 3 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 65 cm ***B.*** 58 cm ***C.*** 57 cm ***D.*** 53 cm ***E.*** 52 cm

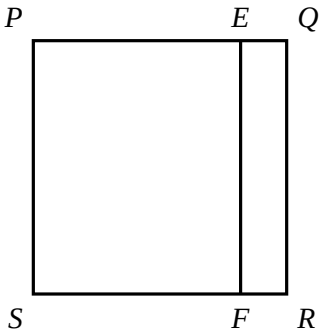
17. This is the outline of a town seen from the river at sunset.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 5

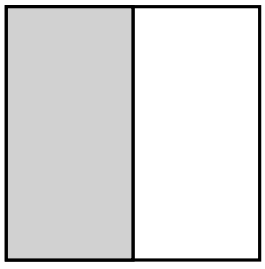
18. The square *PQRS* is cut into two rectangles. The perimeter of *PEFS* is 40 cm and the perimeter of *EQRF* is 26 cm.



How long is the side of the square *PQRS*?

A. 55 cm ***B.*** 33 cm ***C.*** 16 cm ***D.*** 11 cm ***E.*** 10 cm

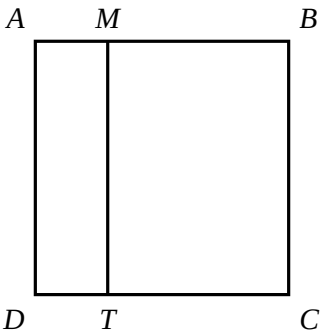
19. Mia cuts a square photo into 2 equal strips as shown. The perimeter of each strip is 60 cm.



What is the perimeter of the square?

A. 20 cm ***B.*** 80 cm ***C.*** 96 cm ***D.*** 100 cm ***E.*** 120 cm

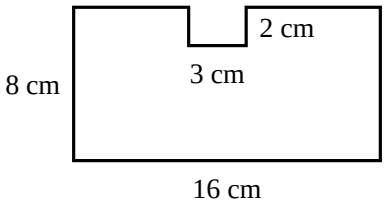
20. The square *ABCD* is cut into two rectangles. The perimeter of *AMTD* is 36 cm and the perimeter of *MBCT* is 48 cm.



How long is the side of the square *ABCD*?

A. 14 cm ***B.*** 21 cm ***C.*** 42 cm ***D.*** 56 cm ***E.*** 70 cm

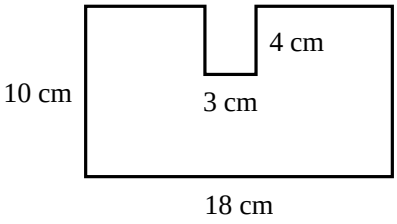
21. A 16 cm by 8 cm rectangle of cardboard has a rectangular notch 3 cm wide and 2 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 45 cm ***B.*** 50 cm ***C.*** 52 cm ***D.*** 54 cm ***E.*** 58 cm

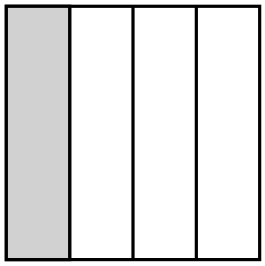
22. A 18 cm by 10 cm rectangle of cardboard has a rectangular notch 3 cm wide and 4 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 64 cm ***B.*** 70 cm ***C.*** 74 cm ***D.*** 79 cm ***E.*** 80 cm

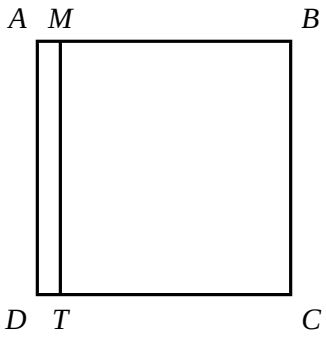
23. Mia cuts a square sheet of paper into 4 equal strips as shown. The perimeter of each strip is 70 cm.



What is the perimeter of the square?

A. 112 cm ***B.*** 122 cm ***C.*** 128 cm ***D.*** 140 cm ***E.*** 280 cm

24. The square *ABCD* is cut into two rectangles. The perimeter of *AMTD* is 24 cm and the perimeter of *MBCT* is 42 cm.



What is the perimeter of the square *ABCD*?

A. 66 cm ***B.*** 55 cm ***C.*** 45 cm ***D.*** 44 cm ***E.*** 11 cm

Answer Key

1.	**C**	**2.**	**E**	**3.**	**E**	**4.**	**A**	**5.**	**A**	**6.**	**C**
7.	**B**	**8.**	**A**	**9.**	**A**	**10.**	**C**	**11.**	**D**	**12.**	**E**
13.	**E**	**14.**	**C**	**15.**	**C**	**16.**	**E**	**17.**	**D**	**18.**	**D**
19.	**B**	**20.**	**A**	**21.**	**C**	**22.**	**A**	**23.**	**A**	**24.**	**D**

Solutions

1. C — All the other pieces can be found in the line, but this one cannot: it has the right shapes but in the wrong order. [2002 Q7 · cannot_belong]

2. E (10 cm) — *MBCT* has sides 15 and 20 cm, perimeter 70 cm. Square: 80 cm. Difference 10 cm (= 2 × 5). [2002 Q6 · square_minus_strip]

3. E — All the other pieces can be found in the line, but this one cannot: its flat part between the two features has the wrong length. [2002 Q7 · cannot_belong]

4. A — Only this piece is found in the line. The others don't fit — B: it is a mirror image of a part of the line; C: it is a mirror image of a part of the line; D: it has the right shapes but in the wrong order; E: its step is shorter than in the line. [2002 Q7 · can_belong]

5. A (46 cm) — The two new edges (3 cm and 3 cm) replace the two removed pieces of the outer edge, which were also 3 cm and 3 cm. So the perimeter is the same as the rectangle's: 2 × (17 + 6) = 46 cm. [2002 Q6 · corner_cut]

6. C (44 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (16 + 6) = 44 cm. [2002 Q6 · corner_cut]

7. B — Only this piece is found in the line. The others don't fit — A: it has a pointed roof where the line has a round dome; C: it has the right shapes but in the wrong order; D: it steps up where the line steps down; E: it has the right shapes but in the wrong order. [2002 Q7 · can_belong]

8. A (72 m) — Side = 36 ÷ 4 = 9 m. There are 2 cuts; each cut is 9 m long and is counted twice (once for each piece next to it): 36 + 2 × 2 × 9 = 72 m. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

9. A (48 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (14 + 10) = 48 cm. [2002 Q6 · corner_cut]

10. C (80 m) — Side = 40 ÷ 4 = 10 m. There are 2 cuts; each cut is 10 m long and is counted twice (once for each piece next to it): 40 + 2 × 2 × 10 = 80 m. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

11. D — Only this piece is found in the line. The others don't fit — A: it steps down where the line steps up; B: it has the right shapes but in the wrong order; C: it steps up where the line steps down; E: its flat part between the two features has the wrong length. [2002 Q7 · can_belong]

12. E (84 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 56 ÷ 8 = 7, so the side is 3 × 7 = 21 cm and the perimeter is 84 cm. [2002 Q6 · strips_to_square]

13. E (72 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 54 ÷ 6 = 9, so the side is 2 × 9 = 18 cm and the perimeter is 72 cm. [2002 Q6 · strips_to_square]

14. C (3) — Pieces 1, 4, 5 fit. Not: 2 (its step is taller than in the line); 3 (it steps up where the line steps down). [2002 Q7 · how_many_fit]

15. C (2) — Pieces 1, 5 fit. Not: 2 (it is a mirror image of a part of the line); 3 (it has a pointed roof where the line has a round dome); 4 (it steps up where the line steps down). [2002 Q7 · how_many_fit]

16. E (52 cm) — The bottom of the notch (3 cm) replaces the 3 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (14 + 9) + 2 × 3 = 52 cm. [2002 Q6 · notch]

17. D (4) — Pieces 1, 2, 3, 5 fit. Not: 4 (it has a round dome where the line has two small bumps). [2002 Q7 · how_many_fit]

18. D (11 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 40 + 26 = 66 cm = 6 sides. Side = 66 ÷ 6 = 11 cm. [2002 Q6 · two_perimeters]

19. B (80 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 60 ÷ 6 = 10, so the side is 2 × 10 = 20 cm and the perimeter is 80 cm. [2002 Q6 · strips_to_square]

20. A (14 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 36 + 48 = 84 cm = 6 sides. Side = 84 ÷ 6 = 14 cm. [2002 Q6 · two_perimeters]

21. C (52 cm) — The bottom of the notch (3 cm) replaces the 3 cm taken out of the top edge, and the two sides of the notch add 2 × 2 cm. Perimeter = 2 × (16 + 8) + 2 × 2 = 52 cm. [2002 Q6 · notch]

22. A (64 cm) — The bottom of the notch (3 cm) replaces the 3 cm taken out of the top edge, and the two sides of the notch add 2 × 4 cm. Perimeter = 2 × (18 + 10) + 2 × 4 = 64 cm. [2002 Q6 · notch]

23. A (112 cm) — A strip is the full side long and 1/4 of it wide, so its perimeter is side + side + 2 × (side ÷ 4) = 10 × (side ÷ 4). 70 ÷ 10 = 7, so the side is 4 × 7 = 28 cm and the perimeter is 112 cm. [2002 Q6 · strips_to_square]

24. D (44 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 24 + 42 = 66 cm = 6 sides. Side = 66 ÷ 6 = 11 cm, perimeter 4 × 11 = 44 cm. [2002 Q6 · two_perimeters]